



Web Application Clone — Master Page & Button Specification

Complete Record of Routes, Pages, Buttons, Exact Connections,
and Visual Styling

10

Total Pages Created

142+

Total Interactive Buttons

10

Interactive Modals

Author / Developer: Aviral Jain (Aviral0105)

Framework: Next.js 16 (App Router) + TypeScript + Tailwind CSS

Backend API: Python FastAPI + SQLite Relational Schema

Repository: <https://github.com/Aviral0105/duolingo-web-clone>

Documentation Date: October 2026

1. Master Page & Route Overview

The application consists of **10 distinct full-stack pages**, backed by a persistent three-column shell layout and 10 contextual modal dialogs.

PAGE ID	PAGE NAME	ROUTE / URL	BUTTON COUNT	PRIMARY PURPOSE
Page 1	Landing & Placement	/ & /welcome	12	Hero presentation, course picker grid & placement assessment.
Page 2	Learning Path (Home)	/learn	10	S-curve path, radial progress ring, star nodes & Unit banner.
Page 3	Full Lesson Player	/lesson/[id]	14	5 exercise types, audio synth, mistake review loop & drawers.
Page 4	Celebration & Rewards	/lesson/[id]/celebrate	6	Confetti, XP, Hindi score, campfire streak & chest claim.
Page 5	Letters / Hindi Alphabet	/characters (/letters)	46	42 Devanagari character cards with native audio synth & quiz.
Page 6	Leaderboards League	/leaderboard	17	Bronze League table, user rank & 12-emoji status picker.
Page 7	Quests & Daily Rewards	/quests	5	Purple Welcome banner, daily XP quests & Monthly challenge.
Page 8	Shop & Super Duolingo	/shop	6	Hearts refills, freeze equip, and Super Duolingo trial.
Page 9	Learner Profile	/profile	10	Avatar hero, stats matrix, Following/Followers hub & Add friends.
Page 10	Settings & Account	/settings/account	11	Sound & animation toggles, profile inputs & logout.
Shell	Persistent Left Sidebar	(All Main Routes)	14	Nav pills, MORE popover menu (6 sub-items), Super shortcut.

PAGE ID	PAGE NAME	ROUTE / URL	BUTTON COUNT	PRIMARY PURPOSE
Topbar	Persistent Stat Bar	(All Main Routes)	4	Course flag (5), Streak (🔥 1), Gems (💎 505), Hearts (❤️ 4).

2. Exhaustive Button Directory & Connections

Global Shell: Left Sidebar (14 Buttons)

BUTTON / LABEL	VISUAL STYLING	WHERE IT IS CONNECTED	ACTION & TARGET STATE
duolingo Brand Logo	Green bold text #58cc02	/learn	Navigates to the active Learning Path home.
LEARN	Cottage icon 🏠, Green active border	/learn	Navigates to Learning Path with unit banners.
LETTERS	Devanagari क icon, Blue border	/characters	Navigates to Hindi alphabet & phonetics roadmap.
LEADERBOARDS	Shield icon 🛡️, Yellow active border	/leaderboard	Navigates to Bronze League rankings and emoji status.
QUESTS	Chest icon 📦, Orange active border	/quests	Navigates to Daily Quests & Monthly challenge hub.
SHOP	Store awning 🏪, Red active border	/shop	Navigates to Duolingo Shop & Power-ups store.
PROFILE	Circular badge A, Cyan border	/profile	Navigates to Learner Profile, statistics, and social hub.
... MORE	Purple circle ...	Sidebar Popover Card	Toggles anchored floating card with 6 sub-actions.
<i>Popover: Settings</i>	⚙️ Gray gear icon	/settings/account	Navigates to Account & Sound Settings.
<i>Popover: Schools</i>	🏫 Sky blue cap	Schools Modal Dialog	Opens Duolingo for Schools classroom portal.
<i>Popover: Podcast</i>	🎧 Amber headphones	Podcasts Dialog	Opens Audio Stories & Podcasts hub.
<i>Popover: Dictionary</i>	📖 Emerald open book	Dictionary Dialog	Opens Hindi vocabulary lookup.
<i>Popover: Help</i>	❓ Indigo question mark	Help Center Dialog	Opens FAQ and bug reporting modal.
<i>Popover: Log out</i>	🚪 Red exit icon	Sign-out Handler	Prompts session logout and redirects to /welcome .

BUTTON / LABEL	VISUAL STYLING	WHERE IT IS CONNECTED	ACTION & TARGET STATE
 Super Duolingo	Purple button <code>bg-purple-50</code>	<code>/shop</code> or Modal	Opens Super comparison sheet with Astronaut Duo.

Global Shell: Topbar Stat Header (4 Buttons)

BUTTON / LABEL	VISUAL STYLING	WHERE IT IS CONNECTED	ACTION & TARGET STATE
Course Flag (IN 5)	Indian flag + score level 5	<code>CourseModal</code>	Opens Course Switcher, <code>+ Add a new course</code> , and <code>+ Add section</code> .
Streak Counter (🔥 1)	Orange flame <code>#ff9600</code>	<code>StreakModal</code>	Opens October 2026 Streak calendar and freeze protection status.
Gems Counter (💎 505)	Cyan diamond <code>#1cb0f6</code>	<code>ShopModal</code>	Opens quick gem balance dialog with direct purchase links.
Hearts Counter (❤️ 4)	Red heart / Super lightning	<code>EnergyModal</code>	Opens Refill for 350 gems and Unlimited Super trial activation.

Page 1: Landing & Course Placement (/ & /welcome) — 12 Buttons

BUTTON / LABEL	VISUAL STYLING	WHERE IT IS CONNECTED	ACTION & TARGET STATE
GET STARTED	Green 3D Button #58cc02	/welcome/course-picker	Navigates to multi-language course selector.
I ALREADY HAVE AN ACCOUNT	White 3D Button border-2	Login Modal	Opens existing credentials login sheet.
Language Grid (7 Cards)	Flags (ES FR DE IN JP 👤 +)	Course Selection State	Selects target curriculum (Hindi selected by default).
CONTINUE (Picker)	Green 3D Button	Placement Assessment	Advances to learner proficiency questionnaire.
Start from scratch	Radio Pill Choice	/lesson/1	Boots immediately into Lesson 1 (Basic Greetings).
Find my level	Radio Pill Choice	Placement Test Flow	Launches diagnostic placement exam.

Page 2: Learning Path (/learn) — 10 Buttons

BUTTON / LABEL	VISUAL STYLING	WHERE IT IS CONNECTED	ACTION & TARGET STATE
GUIDEBOOK (📖)	Transparent pill with book icon	/guidebook/1	Opens Section 1 Unit 1 Grammar & Vocabulary notes.
Active Star Node (★ 1)	Green 3D Node + radial ring	StartLessonModal	Opens level popover with START (+10 XP) and ✕ close.
Locked Node 2 (★)	Gray 3D circle	LockedModal	Displays "Complete previous lesson to unlock".
Locked Node 3 (🎧 Audio)	Gray circle with headphones	LockedModal	Displays listening lesson lock requirements.
Locked Node 4 (📖 Story)	Gray circle with open book	LockedModal	Displays interactive story lock requirements.
Treasure Chest (🏆)	Gold milestone chest	Chest Milestone Modal	Displays gem bounty unlock target.
JUMP HERE? (▶▶)	Purple circular button	Placement Jump Test	Allows skipping Unit 1 via challenge quiz.
Scroll to Top (⬆)	Floating circular white button	Window Viewport	Smoothly scrolls learning path to top banner.

BUTTON / LABEL	VISUAL STYLING	WHERE IT IS CONNECTED	ACTION & TARGET STATE
Super Promo: TRY FOR FREE	White button on dark purple	/shop	Navigates to Super Duolingo activation.
Daily Quests: VIEW ALL	Cyan text #1cb0f6	/quests	Navigates to Quests & Monthly challenges.

Page 3: Full Lesson Player Engine (`/lesson/[id]`) — 14 Buttons

BUTTON / LABEL	VISUAL STYLING	WHERE IT IS CONNECTED	ACTION & TARGET STATE
× Close Button	Gray cross top-left	Exit Confirmation Dialog	Prompts "Quit Lesson" vs "Keep Learning" .
Speaker Normal (🔊)	Blue circular speaker button	Web Audio Synth (1.0x)	Synthesizes target sentence audio at natural speed.
Speaker Slow (🐢)	Turtle blue circular button	Web Audio Synth (0.6x)	Synthesizes target sentence audio at slowed speed.
Choice Cards (1, 2, 3)	White 3D cards with keyboard shortcuts	Exercise Selection State	Selects candidate answer in Multiple Choice.
Word Bank Chips (Pool)	White rounded-2xl chips	Sentence Assembly Slot	Moves clicked word chip into the user response.
Slotted Word Chips	Slotted rounded-2xl chips	Word Bank Pool	Returns slotted word chip back to available pool.
CHECK Button	Green 3D button (Gray disabled)	Answer Evaluator	Evaluates input, triggers chime, and opens Feedback Drawer.
CONTINUE Button	Green (Correct) / Red (Wrong) 3D	Next Step Engine	Advances to next question or triggers Mistake Review Loop.
TOO EASY Pill	Subtle gray pill button	Feedback Dispatcher	Logs difficulty telemetry for adaptive curriculum.
TOO DIFFICULT Pill	Subtle gray pill button	Feedback Dispatcher	Logs difficulty telemetry for adaptive curriculum.
REPORT (🚩)	Flag icon pill button	<code>ReportModal</code>	Opens bug reporting dialog (Audio/Translation issue).
QUIT LESSON	Red outline button in exit modal	<code>/learn</code>	Aborts lesson session and returns to path.
KEEP LEARNING	Green 3D button in exit modal	Active Exercise	Dismisses modal and resumes current question.

Page 4: Lesson Celebration & Rewards (`/lesson/[id]/celebrate`) — 6 Buttons

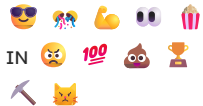
BUTTON / LABEL	VISUAL STYLING	WHERE IT IS CONNECTED	ACTION & TARGET STATE
CONTINUE (XP Screen)	Green 3D Button	Score Milestone Screen	Celebrates XP (+14) and accuracy percentage with confetti.

BUTTON / LABEL	VISUAL STYLING	WHERE IT IS CONNECTED	ACTION & TARGET STATE
CONTINUE (Score Screen)	Green 3D Button	Campfire Streak Screen	Displays Duolingo Hindi Score milestone (+5).
CONTINUE (Streak Screen)	Green 3D Button	Streak Goal Screen	Displays animated campfire flame & "Day 2 starts tomorrow".
Goal Selector (7/14/30/50)	4 Selectable Option Cards	Commitment State	Selects personal streak commitment target.
I CAN DO IT!	Green 3D Button	Chest Reward Screen	Commits streak goal and opens quest reward screen.
CLAIM REWARD & FINISH	Green 3D Button	/learn	Credits 5 gems, updates progress ring, and returns to path.

Page 5: Letters / Hindi Alphabet (/characters) — 46 Buttons

BUTTON / LABEL	VISUAL STYLING	WHERE IT IS CONNECTED	ACTION & TARGET STATE
LEARN THE LETTERS	White 3D Button on cyan	Practice Quiz Modal	Opens phonetic listening practice modal.
Filter: All Letters (42)	Pill Tab (Active: Blue)	Filter State	Displays complete set of 42 Devanagari characters.
Filter: Vowels / स्वर (12)	Pill Tab	Filter State	Filters character grid to 12 vowels (अ to इः).
Filter: Consonants / व्यंजन (30)	Pill Tab	Filter State	Filters character grid to 30 consonants (क to ह).
Character Cards (42 Cards)	White cards, border-2, 🔊 icon, 3-bar green mastery	Web Audio Synth (hi-IN)	Pronounces character aloud and pulses active border.
Quiz: Audio Prompt (🔊)	Large cyan speaker button	Audio Speech Synth	Replays phonetic audio challenge.
Quiz: Choices (4 Cards)	Devanagari letter cards	Quiz Validator	Evaluates matching phonetic character choice.
Quiz: Exit Practice	Gray outline button	Letters Grid	Dismisses practice modal.

Page 6: Leaderboards League (/leaderboard) — 17 Buttons

BUTTON / LABEL	VISUAL STYLING	WHERE IT IS CONNECTED	ACTION & TARGET STATE
START A LESSON	Green 3D Button	/lesson/1	Appears in locked league view to earn prerequisite XP.
Top 3 Learner Rows (3)	Rank cards with Gold/Silver/Bronze medals	Rank Details	Nitheesh Kumar B (#1), AVIRAL JAIN (#2), Seyit Musevi (#3).
Emoji Status Grid (12 Buttons)		Status State Dispatcher	Updates user leaderboard avatar sticker with instant feedback.
League Info Tooltip	Info button	Promotion Dialog	Explains top 10 promotion zone to Silver League.

Page 7: Quests & Daily Rewards (/quests) — 5 Buttons

BUTTON / LABEL	VISUAL STYLING	WHERE IT IS CONNECTED	ACTION & TARGET STATE
Active Quest: Chest (🏆)	Gold interactive chest	Reward Claimer	Claims completed 10 XP quest rewards.
Locked Quest Card	Gray card with padlock 🔒	Quest Details	Shows *"More quests unlock soon"*.
START A LESSON (Right Rail)	White 3D button with blue text	/lesson/1	Launches active lesson to contribute toward monthly badges.
Footer Links (8 Links)	Gray uppercase text	Informational Pages	About, Blog, Store, Efficacy, Careers, Investors, Terms, Privacy.

Page 8: Shop & Super Duolingo (/shop) — 6 Buttons

BUTTON / LABEL	VISUAL STYLING	WHERE IT IS CONNECTED	ACTION & TARGET STATE
START MY 1 MONTH FREE TRIAL	White 3D button on purple gradient	Super Comparison Modal	Opens astronaut Duo Super comparison sheet.
REFILL HEARTS (💎 350)	Blue 3D button	Backend refillHearts()	Deducts 350 gems, refills user hearts to 5.
GET UNLIMITED HEARTS	Purple 3D button	Super Trial Activator	Activates Super mode with unlimited hearts (∞).
EQUIP STREAK FREEZE (💎 200)	Cyan button (2/2 equipped)	Streak Protection State	Equips streak freeze to prevent loss on missed days.
Super Modal: START FREE TRIAL	Purple full-width 3D button	Super State Dispatcher	Activates Super status with celebratory animation.
Super Modal: ✕ Close	Top-right gray close button	Shop View	Dismisses comparison dialog.

Page 9: Learner Profile (/profile) — 10 Buttons

BUTTON / LABEL	VISUAL STYLING	WHERE IT IS CONNECTED	ACTION & TARGET STATE
Avatar Silhouette (+)	Pastel blue card with cyan dashed border and white 	Avatar Creator Modal	Opens avatar builder to customize hair, clothing & skin tone.
Pencil Edit Button ()	White square rounded-2xl top-right	Profile / Avatar Edit Modal	Opens profile editor dialog.
0 Following Link	Sky blue bold text #1cb0f6	Right Rail: FOLLOWING Tab	Switches right rail social hub to Following view.
0 Followers Link	Sky blue bold text #1cb0f6	Right Rail: FOLLOWERS Tab	Switches right rail social hub to Followers view.
Achievements: VIEW ALL	Cyan uppercase bold text	All Achievements Directory	Expands all 15+ Duolingo badges (Wildfire, Sage, Scholar, etc.).
Tab: FOLLOWING	Active cyan underline #1cb0f6	Following Social Hub	Displays ensemble character art & "Learning is more fun when you connect"*.
Tab: FOLLOWERS	Inactive gray text	Followers Social Hub	Displays followers referral link and study buddy prompts.
Find friends Row	 icon + Chevron 	Find Friends Modal	Opens username & contact search dialog.
Invite friends Row	 Green icon + Chevron 	Clipboard Link Dispatcher	Copies referral link with floating success toast.
Avatar Modal: Save Avatar	Green 3D Button	User Profile State	Commits customized avatar and displays toast.

Page 10: Settings & Account Preferences (/settings/account) — 11 Buttons

BUTTON / LABEL	VISUAL STYLING	WHERE IT IS CONNECTED	ACTION & TARGET STATE
Toggle: Sound Effects	Green iOS/Duolingo pill switch	Audio Player State	Enables/disables answer chimes and audio cues.
Toggle: Animations	Green iOS/Duolingo pill switch	UI Animation State	Enables/disables confetti explosions and mascot animations.

BUTTON / LABEL	VISUAL STYLING	WHERE IT IS CONNECTED	ACTION & TARGET STATE
Toggle: Motivational Messages	Green iOS/Duolingo pill switch	Lesson Player State	Enables/disables Duo peeking encouragement interstitials.
Toggle: Listening Exercises	Green iOS/Duolingo pill switch	Curriculum Filter	Includes or excludes audio-listening questions.
SAVE CHANGES	Green 3D Button #58cc02	Settings State Dispatcher	Saves updated preferences and triggers floating green toast.
CANCEL	White 3D Button border-2	/learn	Discards changes and returns to path.
Nav: Account	Active blue item #ddf4ff	/settings/account	Displays account username and email credentials.
Nav: Sound & Audio	Gray item + speaker icon	Sound Settings Anchor	Scrolls to audio toggles section.
Nav: Notifications	Gray item + bell icon	Notifications Settings	Configures streak reminder notifications.
Nav: Super Duolingo	Gray item + purple sparkle icon	/shop	Navigates to Super subscription management.
LOG OUT Button	Red outline card button	Logout Handler	Signs user out and returns to welcome screen.

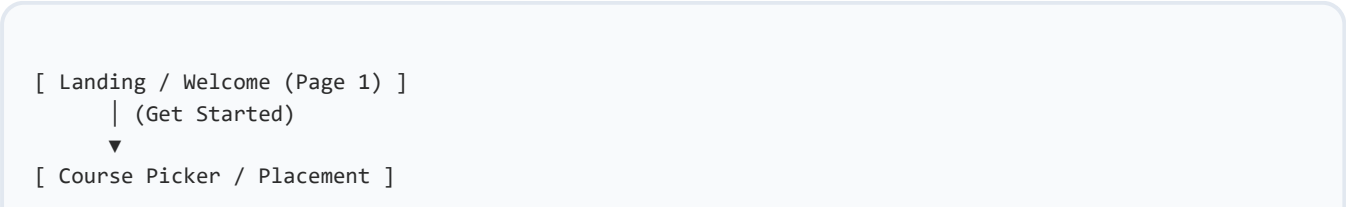
3. Global Interactive Modals (10 Modals)

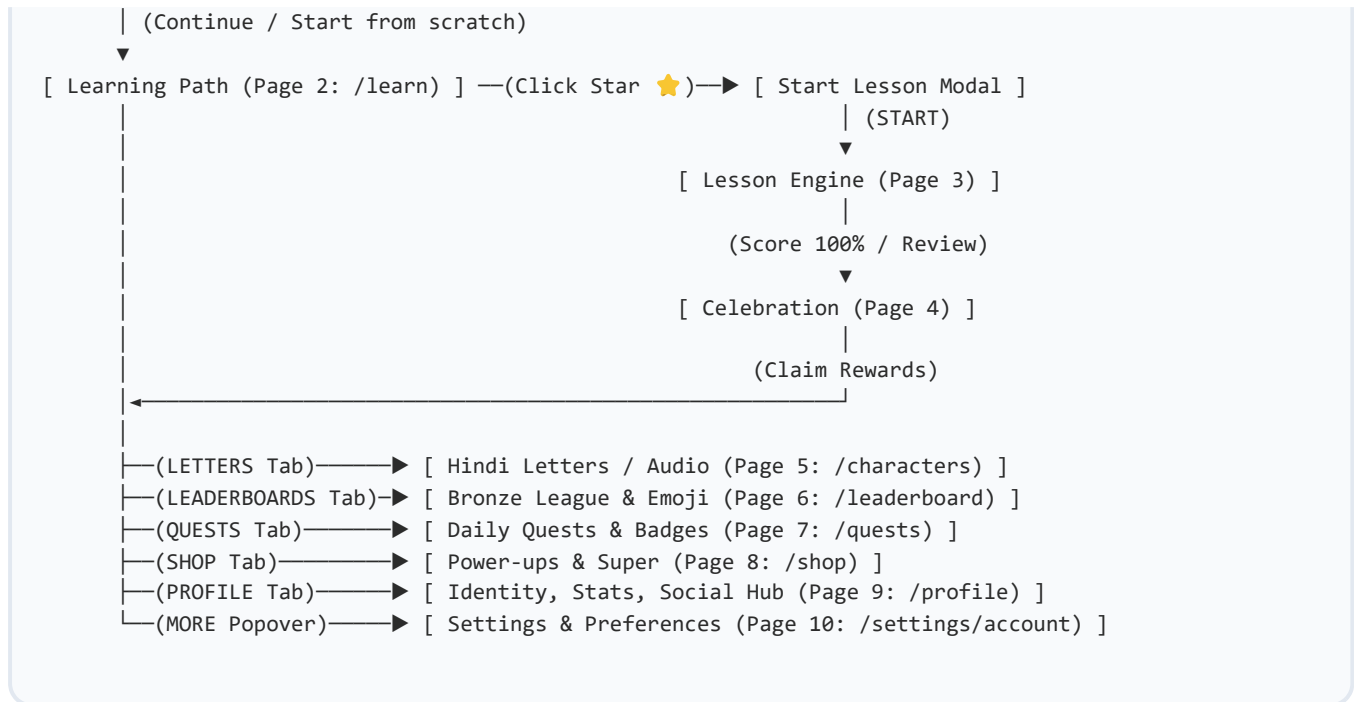
The application implements 10 specialized modal dialogs that overlay the page without full reloads:

MODAL NAME	TRIGGER SOURCE	INTERNAL BUTTONS	BEHAVIOR & ACTION
CourseModal	Topbar flag (IN 5)	Switch Hindi, Switch French, + Add a new course , + Add section , X	Allows course switching, language additions, and curriculum expansion.
StreakModal	Topbar flame (🔥 1)	Calendar navigation, EQUIP STREAK FREEZE , X	Visualizes October 2026 calendar, streak history, and freeze protections.
EnergyModal	Topbar heart (❤️ 4)	Refill (💎 350) , Get Unlimited , X	Restores hearts with gems or starts Super trial.
ShopModal	Topbar gem (💎 505)	Shop navigation shortcut, X	Quick gem overview dialog.
StartLessonModal	Path star node (⭐)	START (+10 XP) , Back to path , X	Boots into /lesson/1 or dismisses on backdrop/Escape.
ExitLessonModal	Lesson header X	QUIT LESSON , KEEP LEARNING	Guards against accidental loss of lesson progress.
ReportModal	Lesson drawer flag (🚩)	Checkboxes, SUBMIT REPORT , CANCEL	Submits question error reports to backend.
LettersQuizModal	Letters LEARN THE LETTERS	🔊 Replay Audio, 4 Letter choices, Exit	Interactive phonetic character matching quiz.
AvatarModal	Profile silhouette / pencil	Color choices, Save Avatar , Cancel	Customizes avatar hairstyle and clothing.
FindFriendsModal	Profile Find friends	Search input, Done , X	Searches user accounts by handle and name.

4. System Connection & Routing Matrix

The following diagram outlines the deterministic transitions between all 10 pages:





— End of Duolingo Web Clone Master Page & Button Specification —